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Item 1

Each multiple-choice question carries 5% mark.

A sinusoidal signal in the ____ domain corresponds to impulses in the ___ domain.

- A frequency; phase
- B phase; time
- C time; frequency ✓
- D frequency; time

Which one of the following statements about energy and power signals is ***not valid***?

- A Power signals can exist (i.e., with non-zero magnitude) over an infinite time duration - they are not time limited.
- B Energy signals exist over a finite time period - they are time limited.
- C Every energy signal is non-periodic.
- D Every non-periodic signal is an energy signal. ✓

Which one of the following statements properly describes the modulation process?

- A Amplify an information (modulating) signal to make it suitable for transmission.
- B Increase the bandwidth of an information signal by multiplying it with a carrier signal.
- C Change the characteristics of a carrier signal to encode an information signal. ✓
- D Change the characteristics of an information signal to encode a carrier signal.

Which of the following describes a disadvantage of full amplitude modulation (AM)?

- A Full AM signal can be demodulated using an envelope detector
- B Full AM has a complex modulator design.
- C The carrier signal for full AM must be sinusoidal.
- D Full AM is not efficient in terms of power usage. ✓

Which of the following statements about amplitude modulation (AM) is ***not valid***?

- A AM requires multiplying the information (modulating) signal with a sinusoidal wave.
- B AM shifts the information (modulating) signal in frequency domain.
- C Envelop detection can be used to receive double-sideband-suppressed-carrier AM. ✓
- D AM is susceptible to noise as the AM signal amplitude represents user information.

What is the range of modulation index for which envelop detection is possible for full amplitude modulation (AM)?

- A Less than 1.
- B Between -1 and 1.
- C Between 0 and 1. ✓
- D Greater than 1.

When compared with amplitude modulation (AM), which of the following statements about frequency modulation (FM) is ***valid***?

- A FM is more susceptible to noise, thus poorer reception quality.
- B FM has an infinite number of sidebands on both sides of the carrier frequency. ✓
- C FM receiver requires use of the carrier signal to recover the information (modulating) signal.
- D FM and AM typically use similar bandwidth to transmit a given information (modulating) signal.

To transmit an information (modulating) signal with bandwidth of 2 kHz by narrow-band frequency modulation (NBFM), what is the minimum bandwidth of the NBFM signal?

- A 4 kHz ✓
- B 2 kHz
- C 1 kHz
- D It cannot be determined without knowing the information (modulating) signal.

Which one of the following statements on wideband frequency modulation (WBFM) is ***not valid***?

- A The modulation index for WBFM is much larger than 1.
- B WBFM is used in commercial TV and radio broadcast.
- C Frequency multiplier typically plays no role in generating WBFM signal. ✓
- D WBFM typically requires a transmission bandwidth much larger than that for amplitude modulation (AM).

Item 2

Numerical Question 1

Let the information (modulating) signal $m(t) = 2\cos(100t) + 2\cos(100t + 2\pi/3)$ modulate the sinusoidal carrier $c(t) = 4\cos(4000t)$ to form a full amplitude modulation (AM) signal.

What is the peak amplitude of $m(t)$? Give your answer in numerical value. [6 marks]

1 ☐

Correct answers:

1 2

Obtain and give the modulation index in numerical value. [4 marks]

1 ☐

Correct answers:

1 0.5

What is the power efficiency of the full AM signal in numerical value. [5 marks]

1 ☐

Correct answers:

1 0.11 ± 0.02

Numerical Question 2

Consider a baseband signal with components at the maximum frequency of 20 kHz.

What is the minimum sampling rate **in kHz** to represent this signal by its samples? [4 marks]

1 ☐ kHz

Correct answers:

1 40.0

Let each sample be quantized into 256 levels. How many bits are required to encode each sample? [4 marks]

1 ☐

Correct answers:

1 8

What is the bit rate **(in kilobits per second)** of the digitized signal? [5 marks]

1 ☐ kilobits per second

Correct answers:

1 320

What is the minimum bandwidth of a noiseless communication channel **(in kHz)** required for transmitting the digitized signal? [5 marks]

1 ☐ kHz

Correct answers:

1 20

Numerical Question 3

Consider a frequency modulated (FM) signal $\phi_{FM} = 10\cos(2\pi f_c t + \frac{5\sqrt{3}}{2}\sin(2\pi f_0 t) - \frac{5}{2}\cos(2\pi f_0 t))$ where $f_c = 10^6$ Hz is the carrier frequency and $f_0 = 1500$ Hz. The power of the information (modulating) signal $m(t)$ is equal to $P_m = 2$.

What is the value of the proportionality constant k_f **in unit of** π? (Hint: Consider the instantaneous frequency of the FM signal ϕ_{FM} .) [6 marks]

1 ☐ π

Correct answers:

1 7500.0

What is the maximum magnitude of the information (modulating) signal $m(t)$? [6 marks]

1 ☐

Correct answers:

1 2.0

By considering the maximum frequency deviation, what is the modulation index β for the FM signal? [5 marks]

1 ☐

Correct answers:

1 5.0

Using the Carlson's rule, what is the bandwidth of the FM signal **in kHz**? [5 marks]

1 ☐ kHz

Correct answers:

1 18.0